

Starting an App

A Django *project* is organized as a group of individual *apps* that work together to make the project work as a whole. For now, we'll create one app to do most of our project's work. We'll add another app in Chapter 19 to manage user accounts.

You should leave the development server running in the terminal window you opened earlier. Open a new terminal window (or tab) and navigate to the directory that contains *manage.py*. Activate the virtual environment, and then run the **startapp** command:

```
learning_log$ source ll_env/bin/activate
(ll_env)learning_log$ python manage.py startapp learning_logs
❶ (ll_env)learning_log$ ls
db.sqlite3 learning_logs ll_env ll_project manage.py
❷ (ll_env)learning_log$ ls learning_logs/
__init__.py admin.py apps.py migrations models.py tests.py views.py
```

The command `startapp appname` tells Django to create the infrastructure needed to build an app. When you look in the project directory now, you'll see a new folder called *learning_logs* ❶. Use the **ls** command to see what Django has created ❷. The most important files are *models.py*, *admin.py*, and *views.py*. We'll use *models.py* to define the data we want to manage in our app. We'll look at *admin.py* and *views.py* a little later.

Defining Models

Let's think about our data for a moment. Each user will need to create a number of topics in their learning log. Each entry they make will be tied to a topic, and these entries will be displayed as text. We'll also need to store the timestamp of each entry so we can show users when they made each one.

Open the file *models.py* and look at its existing content:

```
models.py from django.db import models

# Create your models here.
```

A module called *models* is being imported, and we're being invited to create models of our own. A *model* tells Django how to work with the data that will be stored in the app. A model is a class; it has attributes and methods, just like every class we've discussed. Here's the model for the topics users will store:

```
from django.db import models

class Topic(models.Model):
    """A topic the user is learning about."""
    ❶ text = models.CharField(max_length=200)
    ❷ date_added = models.DateTimeField(auto_now_add=True)
```
